Contact

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Top Skills

Communication
Scientific Writing
Virology

Languages

Chinese (Native or Bilingual)
English (Full Professional)

Publications

Single-phage profiling illuminates viral individuality during bacterial cell fate determination

Wenqing Zhu

PhD candidate in Biophysics and Quantitative Biology
Urbana, Illinois, United States

Summary

Blending my background in chemistry and physics, I'm currently immersed in the world of Biophysics as a Ph.D. candidate.

My research passion lies in single-molecule imaging, specifically understanding bacteriophage infection dynamics in host cells, using both super-resolution fluorescence microscopy and computational modeling. I thrive on continuous learning and have developed expertise in fluorescence microscopy, quantitative analysis and mathematical modeling.

Communication is my forte, refined during my time as the student league administrator in college. As a researcher, I enjoy collaborating with peers and mentors to tackle complex biological challenges.

Beyond my academic pursuits, I'm exploring opportunities in both data science and biotechnology. If you are interested in connecting or discussing potential opportunities, please do not hesitate to reach out!

Excited to explore new possibilities and contribute meaningfully to our shared professional landscape!

Experience

University of Illinois Urbana-Champaign

5 years 3 months

Graduate Research Assistant

October 2021 - Present (5 years 1 month)

GRA in the Golding Lab: https://bacteriophysics.web.illinois.edu/

Investigating viral infection mechanisms through three parallel projects:

(1) Characterizing phage DNA morphology and structure post-entry using super-resolution microscopy (STORM and MINFLUX);
(2) Studying Chi phage motility along bacterial flagella via optical trapping;
(3) Examining lambda phage infection and DNA injection dynamics in motile E. coli via optical trapping
(projects 2 & 3 in collaboration with Yann Chemla lab at UIUC).

Organizer and Instructor, NSF QCB Summer School
August 2024 - August 2024 (1 month)

* Led hands-on training sessions for more than 20 graduate students from UIUC and other universities on advanced imaging techniques over three years.
* Designed and taught experimental modules on super-resolution imaging of bacterial cells, optimizing protocols for high-resolution visualization.
* Taught data analysis workflows for super-resolution imaging, enabling students to accurately interpret and utilize imaging results

Graduate Teaching Assistant
August 2021 - December 2022 (1 year 5 months)

2022 Teaching Awards in the School of Chemical Science

- Teaching Assistant for CHEM544 - Statistical Thermodynamics
- Teaching Assistant for CHEM440 - Principles of Physical Chemistry
- Teaching Assistant for PHYS498 EBP - Experimental Biophysics

Nanjing University
3 years 11 months

Student Affairs Coordinator, Youth League Branch
September 2016 - July 2020 (3 years 11 months)

- Coordinated student engagement activities and communication between the School and the university Youth League Committee.
- Planned and executed student events, including campus culture weeks and volunteer initiatives.
- Managed student data and reporting for annual Youth League evaluations for 200+ students.

Undergraduate Research Assistant
December 2017 - June 2019 (1 year 7 months)
Nanjing, Jiangsu, China

Project: Electrochemical Characterization of MnOx/Carbon Nanocomposites
Selected for Nanjing University Undergraduate Innovation and Entrepreneurship Program (¥5,000 grant)

-Designed and optimized the synthesis of MnOx-filled carbon nanocages (CNCs) using an in-situ MgO-template vacuum filling method.
- Tuned synthesis conditions to achieve uniform carbon encapsulation and controlled MnOx particle distribution.
- Characterized the structural and morphological features of the nanocomposites using X-ray diffraction (XRD), transmission electron microscopy (TEM), and X-ray absorption spectroscopy (XAS).
- Constructed and calibrated an electrochemical workstation to evaluate the materials' performance.
- Measured and analyzed key electrochemical parameters including specific capacity, cycling life, and long-term stability.

- Demonstrated enhanced electrochemical properties of the MnOx/C hybrid materials, showing potential for use in energy storage applications.

Vice Director of Media Department, Student Union
September 2017 - June 2018 (10 months)

-Supervised a team of student writers and editors to manage the School’s official new media presence, including content planning, copywriting, and publication.

- Directed media strategy and execution for key public events such as the New Year Gala, student lectures, and campus-wide activities.

- Developed original content series highlighting traditional Chinese culture and the academic community, improving engagement across platforms.

- Coordinated cross-team collaboration for multimedia production, ensuring consistent tone, branding, and timely delivery of promotional materials.

Member, Media Department, Student Union
September 2016 - June 2017 (10 months)

- Contributed articles and creative content for the School’s new media platforms, covering topics such as student life, traditional culture, and departmental events.

- Participated in the planning and promotion of school-wide events by drafting copy, editing posts, and assisting with visual media.

- Assisted in launching seasonal cultural campaigns, including features on regional customs during major Chinese festivals.

University of Michigan
Undergraduate Research Assistant
July 2019 - December 2019 (6 months)
Ann Arbor, Michigan, United States

Project: Electronic Structure Study of Cu(I) Complexes via X-ray Absorption Spectroscopy

- Synthesized and crystallized a series of Cu(I)-pyridine complexes with systematically varied metal–ligand dihedral angles to probe geometric influences on electronic transitions.

- Performed High-Energy Resolution Fluorescence-Detected X-ray Absorption Spectroscopy (HERFD-XAS) experiments at the Advanced Photon Source, Argonne National Laboratory, focusing on Cu 1s–4p electronic transitions.

- Processed and analyzed XAS data using custom MATLAB scripts developed in-house for spectral fitting and interpretation.

- Carried out time-dependent density functional theory (TD-DFT) calculations using ORCA and Gaussian to simulate and assign X-ray absorption features, enabling detailed correlation between experimental and theoretical spectra.

Education

University of Illinois Urbana-Champaign
Graduate minor in Statistics · (July 2021 - 2026)

University of Illinois Urbana-Champaign
Doctor of Philosophy - PhD, Chemistry · (2020 - 2026)

Nanjing University
Bachelor of Science - BS, Chemistry · (2016 - 2020)